

AI and Data Centers Are Set to Consume 44% of U.S. Energy by 2028—Is the Grid Ready?

Posted by u/christophermatar • 0 points • 59 comments

The U.S. is heading toward a serious **energy challenge** by 2028, and it's largely driven by the booming growth of **data centers** and **AI technologies**. Believe it or not, data centers, which only made up a tiny 1.5% of the nation's energy use a few years ago, are projected to surge to a **whopping 44%** of U.S. electricity demand by 2028. This is largely due to AI, which requires massive computing power to run models like ChatGPT, significantly increasing energy consumption (Source [TC] (<https://www.techradar.com/pro/plan-for-ai-datacenter-power-usage-or-face-the-consequences-energy-companies-told>) [Mckinsey] (<https://www.mckinsey.com/industries/private-capital/our-insights/how-data-centers-and-the-energy-sector-can-sate-ais-hunger-for-power>)).

But it's not just data centers. The rise in **electric vehicle adoption** and the push to bring manufacturing back to the U.S. are also driving up energy needs. All this demand means utility companies will need to boost energy generation by **26%** just to keep up - [source] (https://www.theregister.com/2024/10/11/energy_companies_ai_dcs_consultant_report/).

What's wild is, in key areas like Northern Virginia and Dallas, new data centers are already being delayed for **years** because the grid can't handle the load. If we don't make massive investments in infrastructure soon, we could be looking at serious energy shortages by the end of the decade.

Comments

u/[deleted] • 1 points • 2024-10-15 00:13 UTC

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u/Once_Wise • 1 points • 2024-10-14 22:58 UTC

They are high by an order of magnitude or more. Print a number and people will believe it.

u/[deleted] • 1 points • 2024-10-14 22:31 UTC

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u/Sheharizadian • 2 points • 2024-10-14 21:50 UTC

Oh come on, at least read the [report](https://www.bain.com/insights/utilities-must-reinvent-themselves-to-harness-the-ai-driven-data-center-boom/) that the article is referencing. It says that data centers will make up 44% of the US Electricity demand GROWTH, not the total energy demand.

They said that demand is ~3.8k TWh/year and by the end of the decade could reach 4.2k - 4.8k TWh/year, and 44% **of that increase** (0.4k - 1.0k) would be from increases in demand from data centers. But using those numbers, even using the upper estimate, that would only mean that data centers would make up about 10% of the total energy demand.

I checked some other sources and that math roughly lines up. Data centers currently make up less than 5% of electricity demand, and according to the [ERPI](https://www.energy.gov/policy/articles/clean-energy-resources-meet-data-center-electricity-demand#:~:text=The%20Electric%20Power%20Research%20Institute,of%20total%20load%20in%202023), is projected to grow to 9% by the end of the decade.

Absolutely atrocious reporting from TechRadar smdh

u/Sirisian • 1 points • 2024-10-14 21:27 UTC

Rule 11

u/ID-10T_Error • 1 points • 2024-10-14 21:25 UTC

time to start building datacenters in the desert no need for the grid to be ready require them to have there own powersource.

u/L3g3ndary-08 • 1 points • 2024-10-14 21:13 UTC

>What's wild is, in key areas like Northern Virginia and Dallas, new data centers are already being delayed for years because the grid can't handle the load.

Ah, now it makes sense. This is why Texas is joining the US grid. Not because it's the right thing to do, but it's because it's good for business.....

u/Nosrok • 1 points • 2024-10-14 21:08 UTC

I imagine at a certain point it becomes more financially appealing to create their own power either with a nuclear power plant like MSFT has done or with other forms of power generation.

u/CavemanSlevy • 1 points • 2024-10-14 21:05 UTC

Your title is factually incorrect and directly contradicted by your sources. They may consume 44% of newly added capacity, not of the total grid capacity.

With that being said we are currently not in good shape to keep up with the electrification of the economy. We need not only massive new buildouts in power generation, but also massive build outs of the transmission infrastructure needed to carry it.

u/[deleted] • 1 points • 2024-10-14 20:56 UTC

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